Defining essential land degradation and desertification variables for monitoring and assessment

Survey 1's analysis report

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Sample Size: 31 Respondents

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Key highlights

- 77% of experts agree that the framework for Essential Land Degradation Variables is currently “under development”, validating this initiative as a timely and necessary step to structure a fragmented scientific community.
- The survey confirms a dual priority: scientific harmonization (81%) and policy compliance (74%) are the top objectives, yet 55% of respondents simultaneously demand support for local decision making. This strongly endorses the proposed two-tier architecture (“Overview” + “Context-specific”).
- While the initial list was deemed relevant, experts highlighted significant omissions, specifically calling for the integration of soil biodiversity indicators.
- When ranking selection criteria, experts prioritized data availability (4.46/5), informativeness (4.56/5), and unambiguity (4.23/5) over theoretical scalability or alignment with global initiatives, signalling a demand for a lean, measurable, and immediately actionable framework.
- The survey achieved a high level of constructive participation, with 81% of respondents categorized as “Highly Engaged” or “Constructive”, providing detailed feedback and expressing willingness to participate in the next phase of development.

Participant profile & engagement

The survey attracted a diverse group of experts, predominantly from research and academic institutions (52%, mainly specializing in soil sciences and conservation, soil biodiversity, ecology, hydrology, land degradation assessment, and agricultural management), followed by public agencies (23%) and international bodies (13%).

Geographic representation is strong in the Mediterranean basin (Italy, Spain, Greece, Cyprus, Tunisia, Algeria) with additional input from Northern Europe (Switzerland, United Kingdom, Germany) and China

To quantify participant engagement, we developed a composite **Attitude Score** (ranging from 0 to 40) based on behavioural and qualitative indicators derived from the survey responses:

- Response depth (approximated by length): the total character count of open-text answers, rewarding detailed qualitative feedback.
- Consensus and utility: points awarded for validating the proposed framework (e.g., selecting “Useful” for classification schemes and “Agree” for criteria definitions).
- Future implication: additional points was applied to respondents who explicitly agreed to participate in the follow-up survey.

Based on the final scores, respondents were categorized into four profiles:

- Highly Engaged (23%): Provided detailed textual feedback, validated definitions, and committed to follow-up surveys.
- Constructive/Positive (58%): Validated core content and provided substantial comments, though some declined further participation.
- Neutral/Passive (19%): Agreed with core premises but provided minimal text and declined follow-up.
- Negative (0%): No respondents rejected the methodology outright.

Current state of knowledge

All respondents recognize existing EV frameworks (ECVs, EBVs). Some experts also identify a broader spectrum that includes agricultural, urban, and ecosystem service-related variables. This confirms the relevance of applying this proven methodological framework to the field of land degradation.

Only one respondent (3%) considers the status of Land Degradation EVs as “Well established” (*to be nuanced, maybe the respondent hasn’t fully understood the question*). The majority view the field as fragmented, with 77% describing it as “Still under development” or having “Limited consensus”. 16% of respondent express no opinion and

only one believed that the scientific community is not ready for this framework (but demonstrates a positive attitude regarding our approach. *This one is from UNCCD, perhaps highlighting disconnection between scientific output and immediate policy demand?*).

Conclusion: ELDV initiative is perceived as a necessary and timely contribution to filling a methodological gap and structuring a scientific community still seeking consensus.

Usage and objectives

Analysis of how respondents defined Essential Variables and keywords occurrence (Table 1) reveals three dominant themes:

- Monitoring (42%): Focus on measuring, assessing, and quantifying change over time (e.g., "monitor and compare", "measurable", "quantify").
- Fundamental Nature (29%): Emphasis on creating a "minimum set", avoiding redundancy, and identifying "key" indicators.
- Process Understanding (16%): Desire to understand drivers, pressures, and dynamics (e.g., "drivers of degradation", "processes", "state and dynamics").

Table 1 – Key words used to extract key perspective from respondent definitions of Essential land Degradation variables.

category	key words
Monitor – Assess and track degradation	quantif*, eval*, monitor, assess, measur*, param*, indicator, prox*, data, observation, change, time
Decision – Support decision making	support, decision, decision-mak*, action, warning, intervention, manag*, mitigat*, inform*
Process – Understand underlying mechanisms	driv*, pressure, process, dynamic, impact, cause, effect
Standardize – Allow for harmonization and comparison	standard*, compar*, harmoniz*, common, agree, consistent, shar*, feasibl*, reliab*
Systemic approach – Global perspective	framework, holistic, integrat*, global
Fundamental – Avoid redundancy	redundan*, essential, minim*, fundamental, key, complem*, clear

The definition of Essential Variables is predominantly anchored in operational monitoring and methodological rigor (combining for 73% of responses), indicating that experts prioritize the creation of a lean, measurable framework for tracking change over a deep theoretical analysis of degradation processes.

Q1.3 reveals clear trends regarding expectations for what Essential Land Degradation Variables (ELDV) should support:

- Scientific Harmonization: This is now the most widely supported objective, selected by 81% of respondents. The ability to compare data across regions and studies is viewed as the primary foundation for scientific robustness.
- Multidisciplinary Integration: Supported by 77.4%, this objective indicates a desire to facilitate dialogue between soil, climate, water, and biodiversity communities.
- Policy Compliance: Closely following at 74%, this remains a critical driver. Respondents view ELDVs as essential tools for meeting the reporting requirements of international treaties (UNCCD, SDG 15.3).
- Local Decision Support: While still a majority priority, this is the least selected objective at 61%. This suggests that while local action is important, the immediate consensus prioritizes global standardization and policy alignment over localized management tools.

Intersection of Objectives:

- Holistic Approach: 45% of respondents believe ELDVs should support all four propositions simultaneously.
- The Policy-Local Nexus: There is a 55% overlap between respondents demanding support for both Policy Compliance and Local Decision Support. This strong simultaneous demand validates the proposed two-tier architecture: an "Overview" subset for global treaty reporting and a "Context-specific" subset for local land managers.

Conclusion: We must try to develop a consensus definition for ELDV based on these results. The definition should focus on the monitoring and assessing capacity, while emphasizing the comprehensive (minimum set) and standardized nature of the approach. We need to explore how to align policy compliance with support for local decision-making and field realities, to the extent possible.

Classification preferences

Regarding preference for structuring our candidates' variables:

- Thematic domains: Overwhelmingly preferred (90% found it useful).
- Processes: Also highly rated (87% useful).
- DPSIR Framework: Significantly less popular, with more than the half (52%) finding it "Not useful" for this specific application.

Moreover, respondents strongly support a dual-scale strategy:

- Overview: respectively 58% and 42% found this "Useful" or "Probably useful" for standardized comparison.

- Context-Specific: respectively 71% and 29% found this “Useful” or “Probably useful”, highlighting the need for variables that address local specificities.

Conclusion: While structuring ELDV based on thematic domains and processes, the proposed architecture (overview and context-specific) seems to be a good approach to bridge the operational level and support scientific harmonisation and multidisciplinary integration.

Variable gaps

Missing variables

“Only” 36% of respondents indicated that important variables were missing. Respondents highlighted that the soil biodiversity aspect is lacking. Experts call for concrete biological indicators such as fauna, enzymatic activity, and functional microbial diversity. Current framework considered too focused on physical and chemical aspects.

Experts also mention:

- Hydrological dynamics Specific variables related to water stress, soil moisture, and irrigation efficiency are deemed essential, particularly for monitoring arid regions and managing agricultural water use.
- Anthropogenic drivers: integrate indicators reflecting agricultural practices and human-induced pressures to fully understand degradation drivers.

Some other interesting feedback:

- One respondent correctly noted potential collinearity among proposed variables, an issue already acknowledged and under consideration in the framework's refinement.
- Several respondents suggested including indices such as NDVI. It is important to note that these were intentionally excluded from the variable list, as we consider them as indicators or indices rather than fundamental variables. This distinction requires clearer communication in future iterations to avoid conceptual misunderstanding.
- It seems that experts wish to build upon existing standards while enriching them with new dimensions (biology, water). One interesting point raised was that the key variables for land degradation should in fact be selected from among those already listed in the ECVs, EBVs, EAVs, etc.

Variable exclusion

Some variables should be re-evaluated. Several experts consider certain initial variables to be unnecessary or indirect:

- Static Topography: Plan/profile curvature and terrain attributes are often viewed as static factors rather than dynamic degradation variables.
- Event-Specific Metrics: Fire risk and post-fire recovery were flagged as too specific for a general degradation framework by some, though others argued for their inclusion in context-specific subsets.

Conclusion: whether variables are used or not, it is sometimes difficult to identify trends when experts are working on topics that can be very specific and are responding based on their own experience. We will need to refine our list in a second iteration. However, the results of this survey confirm the relevance of our first approach and initial ELDVs candidates, while calling for major revisions.

Criteria for variable selection or ranking

Respondents evaluated eight criteria and their definition for selecting variables (Table 2). The “data availability”, “Informative” and “Unambiguity” criteria received the highest overall approval rates (84%, 90% and 84%, respectively). While agreement on the core concepts was strong (80–90% full agreement for most criteria), consensus never reached 100% for any single definition. Notably, the criteria for “answering global initiatives”, “spatialization” and “evolvability” elicited the most significant reservations, with 13–32% of respondents agreeing with the concept but rejecting the specific wording, suggesting these definitions may be perceived as either too restrictive or insufficiently precise. The rate of explicit disagreement is low and mainly concerns the “global”, “spatialization” and “evolvability” criteria. This may indicate that the proposed definitions are not only imprecise but also may not reflect some of the experts' opinions. Definitions should therefor be clarified with experts.

Table 2 – Approval rate for each proposed criterion.

	Q41_data	Q41_informative	Q41_global	Q41_spatialized	Q41_scalability	Q41_feasability	Q41_unambigu	Q41_evolvabili
I agree with criteria and definition	83.87	90.32	77.42	61.29	80.65	80.65	83.87	70.97
I agree with criteria but not its definition	16.13	9.68	12.90	32.26	16.13	12.90	12.90	16.13
I don't agree	0.00	0.00	6.45	0.00	0.00	6.45	3.23	6.45
No opinion	0.00	0.00	3.23	6.45	3.23	0.00	0.00	6.45

Assigning distinct weights based solely on average scores proves challenging, as all criteria received relatively high ratings (means ranging from 3.20 to 4.56 out of 5; Table 3), reflecting a general consensus on their collective importance. However, a hierarchy emerges when examining the upper tier: data Availability (4.46), informativeness (4.56), unambiguity (4.23), and feasibility (4.04) stand out as the non-negotiable pillars of the framework. Scalability (3.88), spatialization (3.77), alignment with Global Initiatives (3.43) and the evolvability (3.20) of variables are perceived as secondary considerations. This suggests that experts prioritize the immediate operational robustness and clarity of variables over their theoretical scalability or policy alignment, indicating that future weighting schemes should heavily favour practical measurability and data reliability.

Table 3 – Mean and median grade for each proposed criterion. Ranging from 0 (not important at all) to 5 (very important).

	Q43_data	Q43_informativ	Q43_global	Q43_spatialize	Q43_scalabilit	Q43_feasabili	Q43_unambigu	Q43_evolvability
Mean	4.46	4.56	3.43	3.77	3.88	4.04	4.23	3.20
Median	5.00	5.00	4.00	4.00	4.00	5.00	5.00	4.00

Conclusion: It seems that the quality of the data and the information it contains is of paramount importance to the experts. This exercise is important, but the definitions need to be revised, and closer collaboration with experts should allow for more precise weighting.

Strategic Recommendations

Based on the qualitative feedback and quantitative scoring, the following actions are recommended for the next phase of framework development:

- Strengthen the 2-tiered structure: define what variables are the basis (e.g.: remote sensing based, non-categorical/without subjective ranking, …) without discarding widely used variables that are relevant at local scales and require more precise data.
- Revise the content of the candidate list, by including more precise variables related to biodiversity. Conduct a colinear analysis to discard too correlated variables.
- Refine Definitions and methodology: Clarify variables' definition and measurement unit, the distinction between variables and indicators, and the criteria's definition.

Conclusion

The survey results confirm that the scientific community is ready to transition from conceptual development to operational definition. The consensus is clear: experts demand a pragmatic, non-redundant monitoring framework anchored in data availability and scientific rigor, rather than a theoretical exercise. While the primary mandate is to harmonize global science and satisfy international policy obligations (SDG 15.3, UNCCD), the strong simultaneous demand for local relevance validates the proposed dual-tier architecture. By balancing a standardized "Overview" core for global reporting with a flexible "Context-specific" subset for local management, the Essential Land Degradation Variables framework can effectively bridge the gap between high-level treaty compliance and on-the-ground decision-making. The immediate next steps involve refining variable definitions, integrating critical biological and hydrological gaps, and resolving conceptual ambiguities to finalize a robust, actionable tool for the global community.